



CONTACT

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TECHNICAL SKILLS

CONTROL & GNC

Flight Dynamics, Flight Control, Guidance & Navigation, Adaptive Control, PID/MRAC, 6-DOF Dynamics

PROGRAMMING

Python, C, MATLAB/Simulink, Machine Learning, ROS/Gazebo, TLA+, Git/SVN

SYSTEMS & TOOLS

Systems Engineering, Modelling & Simulation, CATIA, CFD, STK, Ansys HFSS

CERTIFICATION

Systems Tool Kit (STK)- Level 1 (Ansys)

EDX – 16.00x: Introduction to Aerospace Engineering: Astronautics and Human Space Flight (Massachusetts Institute of Technology)

SOFT SKILLS

Curiosity
Rigorous Thinking
Adaptability
Teamwork
Problem Solving

LANGUAGES

English	Native (C2)
French	Intermediate (B1)
Hindi	Native (C2)

ACTIVITIES

Vice President - Outgoing Global Exchange at AIESEC Carleton

Lead a team of exchange managers facilitating exchanges for internship and volunteer opportunities for participants.

ANUSHA JAIN

ISAE-SUPAERO - MSc Aerospace Engineering | Systems & Control
Flight Control • Guidance & Navigation • Aerospace Software

Final-year MSc student at ISAE-SUPAERO seeking a 6-month end-of-studies internship from April 2027. Interested in flight control, guidance and navigation, and in bridging aerospace engineering with software to develop autonomous and reliable systems.

EXPERIENCE

 **ISAE-SUPAERO – Graduate Student Researcher | Glider Control & RL** Feb'26 - Present
Toulouse, France

Atmosphere-Aware RL Controller for High-Altitude Glider Flight

- Developing a 6DOF glider simulator in MATLAB/Simulink with atmospheric turbulence models.
- Designing an adaptive RL controller, benchmarked against a PID baseline (in progress).



SupaeroMoon Sep'25 - Present
Toulouse, France

Head of Systems Engineering

- Leading systems engineering across three rover projects (ERC 2026, MDRS, open-source modular rover).
- Defining architecture and interfaces between autonomy, structures and science subsystems; tracking compliance with mission requirements.



BlackBerry QNX – CO-OP May'23 - Aug'23
Ottawa, Canada

Safety Systems Software Developer

- Drafted and revised safety manuals for Automotive OEM and RTOS products.
- Automated code traceability (Python, C, JSON) and supported HARA/HAZOP safety assessments using TLA+ formal modelling.

ACADEMIC PROJECTS

UAV Flight Control – Adaptive Control Apr'26 - Jun'26

- Designed a cascade PID baseline, then implemented a 3-axis MRAC (MIT rule) altitude controller for a quadrotor under sudden mass drop.
- Validated in Simulink simulation and physical flight tests.

Airbus Fly Your Ideas Challenge Mar'26 - Jun'26

- Led a multidisciplinary team to Round 2 applying post-quantum cryptography to aerospace cybersecurity.
- Directed technical development, system design, and team coordination with mentors and teammates.

Cooperative UAV Simulation & Control – ROS/Gazebo Sep'24 - Apr'25

- Integrated and simulated a dual-drone cooperative payload system using ROS, MAVROS, ArduCopter, and Gazebo.
- Led project integration, aligning system architecture, control design, and simulation.

Spacecraft Attitude Dynamics and Control Sep'24 - Dec'24

- Modelled spacecraft attitude dynamics and reaction-wheel momentum exchange in MATLAB/Simulink with quaternion feedback control.
- Built an orbit propagator and modelled four perturbation torques (SRP, drag, gravity gradient, magnetic) to validate attitude response.

EDUCATION

 **MSc in Aerospace Engineering, Aerospace Systems and Control** 2025 - 2027
Toulouse, France

ISAE SUPAERO

- Space Propulsion, Adaptive Control, Artificial Intelligence for Control, Multiple-Input Multiple- Output (MIMO) Systems, Hybrid Control, Robust and Optimal Control.



B.Eng. in Aerospace, Space Systems Design 2021 - 2025
Carleton University Ottawa, Canada

- Transatmospheric Space Propulsion, Orbital Mechanics, Spacecraft Attitude Dynamics and Control, Spacecraft Design, Heat Transfer for Aerospace Applications.